Year 8 SEAL Science
Energy Practice Test – Medium

Details	
Student Name:	Teacher Name:
Date Set: ____/____/26	Due Date: ____/____/26

Curriculum Assessed

On completion of this unit the student should be able to classify energy as kinetic or potential, calculate kinetic and gravitational potential energy, explain heat transfer by conduction, convection and radiation, analyse energy transfers and transformations using efficiency and Sankey diagrams, calculate appliance running costs, and evaluate ways to reduce energy use. This practice test assesses Chapter 7 Energy.

Task Overview

Section A – Multiple Choice: 12 questions, 1 mark each (12 marks). Answer on the separate multiple-choice answer sheet.
Section B – Short Answer: 7 scenario-based questions (28 marks). Answer in this booklet.

Conditions

- Test conditions
- Formula sheet provided on page 2
- Permitted materials: pens, pencils, calculator, ruler, sharpener, eraser, highlighters.
- Not permitted: notes of any kind, blank paper, white-out, mobile phones, smart watches or any other electronic device.
- 55 mins writing time

Marking

- 40 marks
- Marked according to teacher-made marking guide.
- Show all working in calculation questions; give every final answer with a unit. When you explain or justify, use the numbers you are given.

A+	A	B+	B	C+	C	D+	D	E+	E	NS
90-100%	80-89%	75-79%	70-74%	65-69%	60-64%	55-59%	50-54%	45-49%	40-44%	<40%

Total Marks:		Percentage:		Letter Grade:	

Formula sheet

Use these formulas and units in Sections A and B.

Quantity	Formula
Kinetic energy	KE = ½ × m × v^2 (m in kg, v in m/s, KE in J)
Gravitational potential energy	GPE = m × g × h (g = 10 N/kg, h in m, GPE in J)
Efficiency	Efficiency = useful energy out ÷ total energy in × 100%
Energy used by an appliance	Energy (Wh) = power (W) × time (h)
Units	1 kWh = 1,000 Wh · 1 kW = 1,000 W
Running cost	Cost ($) = energy used (kWh) × tariff ($ per kWh)

SECTION A – Multiple-choice questions (12 marks)

Answer **all** questions on the separate multiple-choice answer sheet by filling in the letter of the best response. Each question is worth 1 mark. No marks are deducted for incorrect answers. Do **not** write your Section A answers in this booklet.

Question 1

Which energy on a farm is classified correctly?

- **A.** the hot water in the dairy's tank: thermal energy, a potential (stored) type
- **B.** a hay bale resting in the shed loft: gravitational potential energy, a potential (stored) type
- **C.** the hum of the shearing-shed motor: sound energy, a potential (stored) type
- **D.** the diesel in the tractor's tank: chemical energy, a kinetic (movement) type

Question 2

A grain truck of mass 8,000 kg rolls slowly up to the silo. It has 36,000 J of kinetic energy. What is its speed?

- **A.** 3.0 m/s
- **B.** 2.1 m/s
- **C.** 4.5 m/s
- **D.** 9.0 m/s

Question 3

In which of these does energy only move from one object to another, without changing type?

- **A.** A tractor engine burns diesel to turn the tractor's wheels.
- **B.** A solar panel charges the shearing shed's battery.
- **C.** A hot-water bottle warms a newborn lamb in the shed.
- **D.** A log burns in the wood stove and heats the farmhouse kitchen.

Question 4

Two identical bread rolls sit on a bench in a bakery at 20 °C. Roll P is at 90 °C and roll Q is at 50 °C. Which roll loses heat faster during the next minute, and why?

- **A.** Roll Q, because a cooler roll has less heat to hold on to, so it lets heat go more easily.
- **B.** Both lose heat at the same rate, because they are identical rolls in the same room.
- **C.** Roll Q, because cold from the room can flow into a cooler roll more easily.
- **D.** Roll P, because it is much further above room temperature, and a bigger temperature difference makes heat transfer faster.

Question 5

A farm windmill uses the wind to pump water up into a tank on a tall stand. Which chain shows the main energy changes?

- **A.** kinetic (wind) → kinetic (turning blades and pump rod) → gravitational potential (water in the raised tank)
- **B.** kinetic (wind) → electrical (turning blades) → gravitational potential (water in the raised tank)
- **C.** gravitational potential (wind) → kinetic (turning blades) → thermal (water in the raised tank)
- **D.** chemical (wind) → kinetic (turning blades) → kinetic (water resting in the raised tank)

Question 6

A farmer compares three electric pumps that could fill the dairy's water tank. The table shows the energy each pump uses and the useful energy it gives to the water every second.

Which pump is the most efficient?

Pump	Electrical energy in each second (J)	Useful energy to the water each second (J)
P	400	100
Q	500	150
R	1,200	240

A. Pump P, because it wastes the fewest joules each second.

B. Pump R, because it gives the most useful energy each second.

C. Pump Q, because it turns the largest share of its input into useful energy.

D. Pump P, because it uses the least electrical energy each second.

Question 7

Which statement about First Nations fire-making methods is correct?

A. Rubbing wood together creates new energy that was not there before.

B. Woods, tinders and methods were chosen to suit the materials found on local Country.

C. The same method and the same materials were used everywhere in Australia.

D. Percussion works by releasing heat that is stored inside cold stones.

Question 8

The energy label on the dairy's new milk-vat cooler gives its energy use in kilowatt-hours (kWh). What is one kilowatt-hour?

A. the power of an appliance, measured in thousands of watts

B. the number of hours an appliance runs each day

C. the cost of running an appliance for one hour

D. the energy used by a 1,000 W appliance running for one hour

Question 9

An energy audit of the dairy lists three appliances, their power and how long each runs each day.

Which appliance uses the most energy each day?

Appliance	Power (W)	Time on each day (h)
Hot-water heater	3,600	4
Milk-vat cooler	2,000	10
Water pump	750	24

A. the milk-vat cooler

B. the hot-water heater, because it has the highest power

C. the water pump, because it runs for the longest time

D. You can't tell without each appliance's star rating.

Question 10

The owners of a farmhouse are choosing between R1.5 and R4.0 ceiling batts. Which statement is correct?

A. The R4.0 batts let heat through faster, because they hold more heat.

B. R-values only matter in winter, because the only job of insulation is to keep heat in.

C. The R4.0 batts slow heat flow more, so they help keep the house warmer in winter and cooler in summer.

D. The R-value is the number of degrees the batts warm the room by.

Question 11

A grain auger (a long turning screw inside a tube) lifts 500 kg of wheat to the top of a silo. The wheat gains 60,000 J of gravitational potential energy. How high is the top of the silo? Use g = 10 N/kg.

A. 0.083 m

B. 12 m

C. 24 m

D. 120 m

Question 12

The shearers' quarters need a new washing machine. Model X uses 190 kWh per year and Model Y uses 240 kWh per year. Electricity costs 30 c per kWh. How much more does Model Y cost to run each year than Model X?

A. $50.00

B. $1,500.00

C. $129.00

D. $15.00

END OF SECTION A

SECTION B – Short-answer questions (28 marks)

Answer **all** questions in the spaces provided. Show working for all calculations. Give every final answer with a unit.

Question 1 (7 marks)

Hay season. Use g = 10 N/kg.

Part 1 – Carting the hay

A tractor pulling a trailer of hay has a total mass of 5,000 kg. It drives along the farm track at 2.0 m/s.

a) Calculate the kinetic energy of the tractor and trailer.

2 marks

Part 2 – Into the loft

At the shed, an electric bale elevator lifts each bale of mass 24 kg up to a loft 3.5 m above the ground. For each bale, the elevator's motor uses 1,400 J of electrical energy.

b) Calculate the gravitational potential energy gained by one bale.

2 marks

c) Calculate the efficiency of the bale elevator.

1 mark

d) Name two forms the wasted energy takes.

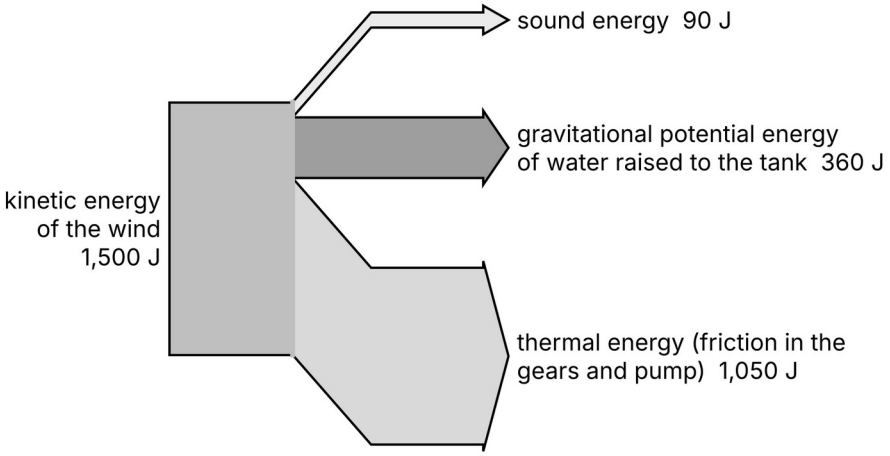
1 mark

e) Suggest one change to the elevator that would raise its efficiency.

1 mark

Question 2 (2 marks)

Windmill water pump. Figure 1 is a Sankey diagram for a farm windmill that pumps water up into a tank. It shows the energy changes every second, before the gears are greased.



The farmer greases the gears and the pump rod. Now, every second, the same 1,500 J of wind energy gives 480 J of gravitational potential energy to the water, 60 J of sound energy and 960 J of thermal energy.

Draw a labelled Sankey diagram for the windmill pump after the gears are greased. Draw it so each arrow's width matches the amount of energy it carries.

2 marks

Question 3 (4 marks)

Dairy hot water. The dairy's hot-water system keeps water at 75 °C. The hot water flows through a 20 m copper pipe to the milking shed. The outside of the pipe feels hot.

a) Explain, in terms of particles, how heat travels from the hot water to the outside surface of the copper pipe.

2 marks

To save energy, the pipe needs lagging (a foam or foil cover). The farmer plans to wrap the pipe tightly in one layer of shiny foil, instead of thick foam lagging.

b) Evaluate the farmer's plan.

Shiny foil mainly reduces …

Foam lagging mainly reduces …

2 marks

Question 4 (3 marks)

Fire plough. At the country town's show, a First Nations ranger demonstrates the fire plough. She pushes the tip of a hard stick quickly up and down a groove in a soft, dry piece of wood. Hot wood dust gathers at the end of the groove and forms a glowing ember. She tips the ember into a tinder bundle and blows gently until it bursts into flame.

a) Describe the energy transformations, from the ranger's muscles to the flame.

2 marks

b) Fire is then used to cook food. Describe one energy transfer that happens when the food cooks.

1 mark

Question 5 (2 marks)

Farmhouse bedroom. In winter, the back bedroom of the farmhouse is always cold. It has a single-glazed window, and a 1 cm gap under its outside door.

a) Cold outside air flows in through the gap under the door. Identify the main type of heat transfer involved.

1 mark

b) Suggest one upgrade for the window, and explain how it reduces heat loss.

1 mark

Question 6 (4 marks)

Country bakery. The bakery's old deck oven has a power of 9,000 W. On a normal day it runs for 6 hours. Electricity costs $0.28 per kWh.

a) Calculate the energy the oven uses in one day, in kWh, and its running cost for one day.

2 marks

b) The baker wants the oven to cost no more than $10.00 a day to run. Calculate the greatest number of whole hours the oven can run each day. Show your working.

2 marks

Question 7 (6 marks)

Investigation. Students at the town's school test this question: *How does the tilt angle of a solar panel affect its power output?* They set up a small solar panel facing north in an open paddock. They tilt it to six different angles (0° is flat on the ground) and record the power output with a meter. They do three trials at each angle, all between 12:00 and 12:30 pm on a clear day.

Tilt angle (°)	Trial 1 (W)	Trial 2 (W)	Trial 3 (W)	Average (W)
0	6.6	6.8	6.7	6.7
15	7.5	7.7	7.6	7.6
30	8.0	5.2	8.2	
45	7.9	7.7	7.8	7.8
60	7.0	7.2	7.1	7.1
75	5.8	6.0	5.9	5.9

a) Identify the independent variable, the dependent variable, and one variable the students kept the same.

Independent variable: _____

Dependent variable: _____

Kept the same: _____

2 marks

b) One result in the table is an outlier. Identify it, and calculate the average power output at 30° that the students should record.

2 marks

c) A student concludes: "The steeper the panel is tilted, the more power it makes." Evaluate this conclusion, using data from the table.

2 marks

END OF SECTION B

Additional space

Clearly number any response written here.

END OF TEST